



António José Freitas Machado Dórdio Gomes

MECHANICAL ENGINEER
AERONAUTICAL STRUCTURES & COMPOSITES

Profile

Mechanical engineer specialized in structural analysis and composite materials, with experience in finite element modelling, bonded joints and experimental validation. Worked on structural design and testing of composite components and mechanical systems through research projects and prototype vehicle development. Strong background in Abaqus-based simulation, multidisciplinary engineering projects and structural performance assessment. Further details and selected engineering projects can be found in the **engineering portfolio**.

Professional Experience

Graduate Researcher — Master's Thesis

Sep 2025 – Mar 2026

Department of Aerospace Science and Technology — Politecnico di Milano, Italy

- Developed **Abaqus/Explicit FE models** for **low-velocity impact damage** in **Glass/PPS laminates**
- Implemented **Hashin intralaminar damage** and **cohesive zone modelling** for fibre fracture and delamination
- Introduced a **bi-phasic shell–solid discretisation** to capture **indentation** and **matrix-dominated deformation**
- Validated simulations with **impact tests** and **ultrasonic C-scan data**, predicting **delamination area within ~15%**

Internship — WEG — Bachelor's Thesis

Feb 2023 – Jun 2023

Product Development and Certification Department

- Developed **ANSYS thermal models** to evaluate **heat dissipation** in electric motor housings
- Quantified **cooling performance differences** between **steel and cast iron motor housings** (up to ~52%)
- Performed **parametric analysis of fin geometry**, recommending **chamfered fins** for steel housings
- Validated simulations with **experimental tests**, achieving **7.7% deviation**

Summer Internship — BI-SILQUE

Aug 2022 – Sep 2022

Maintenance Department

- Supported **maintenance operations** and **industrial machine monitoring** in a manufacturing environment
- Managed **spare parts inventory** and **maintenance records** using **Excel and Access**

Engineering Experience

Formula Student FEUP — Electric Formula Racing Prototype Development

Wishbones & Vibrations Lead

Sep 2024 – Aug 2025

- Led a 2-person sub-team** developing and manufacturing **CFRP suspension wishbones** with bonded **7075-T6 aluminium connectors** for the prototype vehicle
- Performed **Abaqus simulations** of aluminium connectors across **25 suspension load cases** and **adhesive joints** validated through correlation with **tensile tests**
- Manufactured and tested **~40 representative tube–connector joint specimens** in **tensile tests**
- Investigated **adhesive joint failures** and implemented a **bi-adhesive bonding solution** validated up to **80 °C**

Transmission Engineer

Sep 2023 – Sep 2024

- Designed **in-house drivetrain components** (**tripod housing, driveshaft and halfshafts**) for the EV prototype
- Performed **structural and safe-life fatigue analysis** of rotating drivetrain components
- Verified **spline interfaces** and **standards compliance** using **involute spline inspection**
- Contributed to a **4 kg drivetrain subsystem mass reduction** vs previous prototype

Education

MSc in Mechanical Engineering – Aeronautical Structures and Vehicles

2023 – Present

Faculty of Engineering, University of Porto (FEUP)

Mechanical Engineering Bachelor

2020 – 2023

Faculty of Engineering, University of Porto (FEUP)

Awards

- 2025 (EV Class) — Formula Student Germany:** 3rd Cost · 9th Business · 25th Design · 36th Overall — **Presented Cost & Cost Final. Formula Student Portugal:** 1st Business · 2nd Design · 3rd Cost · 4th Overall — **Presented Cost & Cost Final. Formula Student Netherlands:** 1st Business · 3rd Cost.
- 2024 (Concept Class) — Formula Student Switzerland:** 1st Overall · 1st Design · 1st Cost · 1st Business. **Formula Student UK:** 2nd Overall · 1st Cost · 2nd Lap Time · 3rd Design — **Presented Design & Cost Final. Formula Student Portugal:** 1st Overall · 1st Design · 1st Cost · 1st Business — **Presented Cost.**

Academic & Volunteering Activities

NEEM FEUP - Pedagogic Department

Mar 2021 – Jun 2025

Mechanical Engineering Students' Association — FEUP

- Organized **MecGames** (Engineering Competition) and **WelcoMec** (First-Year Mentoring Program)
- Staff member for **Connecting Gears 2022–25** (Mechanical Engineering Technical Conference)
- Staff member for **ENEEM 2023** (National Meeting of Mechanical Engineering Students)

Volunteer Work — Banco Alimentar Contra a Fome do Porto

May 2023

📍 Porto, Portugal
☎ +351 939 290 065
✉ antmachado7@gmail.com
🌐 linkedin.com/in/antmachado7
🌐 Portfolio Website

Certifications

CSWA — Certified SolidWorks Associate, Dassault Systèmes, Aug 2023

Composites Design Workshop XXVII, Prof. Steve Tsai, Stanford University, Jul 2024

Technical Skills

Abaqus

ANSYS

Python

MATLAB

Excel

Inventor

SolidWorks

Fusion 360

LaTeX

Languages

Portuguese	Native
English	C1
Spanish	B1
French	A2

Engineering Focus

Composite Structures
Finite Element Analysis
Structural Mechanics
Impact Damage Mechanics
Bonded Joints